

Math 201-06
Checkpoint 10

Directions:

- **Desmos.** We will often use *Desmos* for computations and graphing, just like we do during class. Any work done in *Desmos* must be summarized and briefly explained in writing on your paper.
 - **Resources.** During each checkpoint, you may access your 1 page of notes, *Desmos*, and our Blackboard page (for submission); no other uses of your computer or tablet are permitted. No collaboration with other human beings or use of other internet resources are allowed.
 - **Competency.** Each week, your goal is to demonstrate competency on as many of the included learning targets as you can. Each question will be marked either “Y” or “NY”: Y for “yes!”, NY for “not yet”. Every learning target will appear on three consecutive checkpoints, so you’ll have multiple attempts to demonstrate competency. “Competency” is defined as work that demonstrates clear and complete understanding of the relevant concepts in the learning target. While the work may include a minor error, or one or two minor steps may not be well explained, otherwise the work is complete, correct, and convincing, including the proper use of notation, and appropriate explanation of meaning that uses correct terminology. (On rare occasions, you might get a “NY*”; such a mark means you are close to competency and that you have the option to have a short conversation with me prior to the next Checkpoint and attempt to demonstrate competency through the conversation aloud.)
 - **Submission.** When finished, use a scanning app to create digital images of your work and upload it to Blackboard under the current Checkpoint location, doing so by 1 pm.
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Learning Targets: For Checkpoint #10, there are three learning target being assessed:

LT 10: I can use implicit differentiation to find $\frac{dy}{dx}$ for a function defined implicitly, using basic derivative and structure rules in the process. (Section 2.7)

LT 11: I can solve related rates problems and interpret the results in the given context. (Section 3.5)

LT 12: I can solve applied optimization problems using appropriate derivative tests and write a concluding sentence that justifies my work. (Section 3.4)

To demonstrate competency on Learning Target # n , you need to correctly respond to all or nearly all of the prompts in question # n . If you have already passed a learning target on a prior Checkpoint, you don't need to re-attempt it on this one.

10. Consider the curve defined implicitly by the equation

$$y^3 - 5xy + 7x^3 = 3$$

- (a) Use implicit differentiation to find an expression for $\frac{dy}{dx}$ that depends on x and y . Show all of your work and thinking clearly, using correct notation.

- (b) Notice that the point $(1, 1)$ lies on the curve $y^3 - 5xy + 7x^3 = 3$. Find the exact slope of the tangent line to curve at $(1, 1)$. Show clearly how you determined this value, and label your work with proper notation.

11. At a 4-way intersection, there's a gas station 1 mile north of the intersection. An automobile traveling east-west passes through the intersection traveling east, missing the turn to the gas station.

If the automobile is traveling 50 miles per hour when it is 2 miles east of the intersection, how fast is the distance* between the car and the gas station increasing at this instance?

*(By "the distance", we mean "the distance if the car could drive in a straight line to the gas station", not by retracing its path back to the intersection).

To earn a Y on this question, you must fully justify your answer using calculus, including

- *a clear, labeled diagram with variables identified*
- *appropriate equations that relate changing quantities and their rates of change*
- *proper notation involving derivatives*
- *a clear summary of your result, with correct units*

12. Suppose that we want to construct an open box (no top) with a square base that has the following properties: the material for the base costs \$3 per square foot, the material for the sides costs \$2 per square foot, and we want to build the box that holds the most volume for a cost of \$10.

Your task is to fully *set up* an optimization problem to be solved on an appropriate domain. *To earn a Y on this problem, you must fully justify your answer using calculus, including*

- *Draw a picture of the situation and introduce appropriate variables.*
- *Identify a constraint equation for how variables are related.*
- *State the quantity being optimized and find a formula that represents this quantity as a function of a single variable.*
- *Determine, with a sentence of justification, the domain on which the function should be optimized.*
- *Write 1-2 sentences to explain what you would do from here (without actually doing so) to use calculus to find the optimal box. In particular, be sure to explain how you would determine whether you had found an absolute maximum or absolute minimum.*